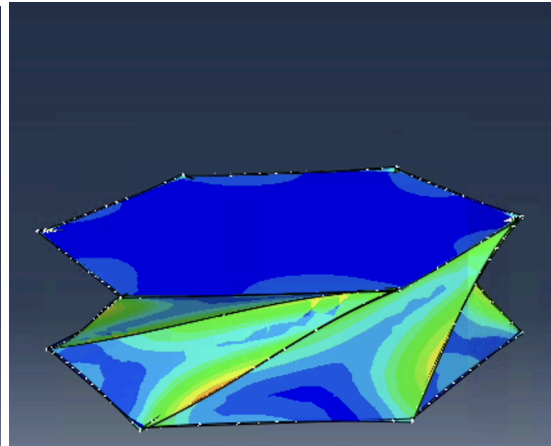
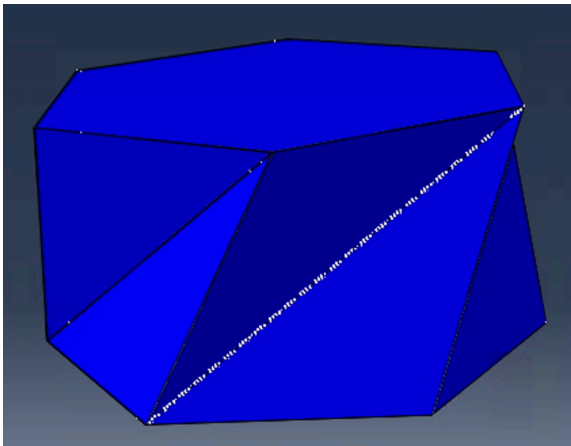
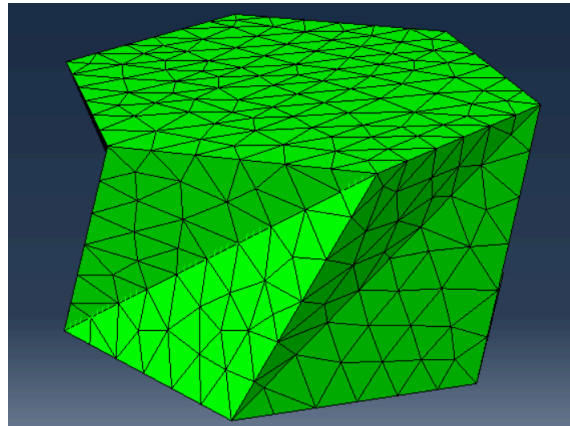




ES-128 Final Project

“Kresling Unit Cell for Abaqus CAE”

Anna Burgess and Ella McRitchie

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Abstract:

An accurate finite element model of a Kresling origami unit cell was developed in Abaqus CAE to replicate the mechanical response of a physically fabricated structure. The modeling process incorporated detailed geometric parameterization, material selection, and hinge representation to capture the nonlinear deformation behavior observed under axial compression. The fabrication process and experimental validation using Instron compression testing informed the calibration of torsional spring stiffness values within the simulation. Parametric sweeps were conducted across key geometric variables, including unit cell height and polygon side count, to evaluate sensitivity in stiffness and force–displacement response. The results demonstrate that the Kresling pattern exhibits strong dependence on geometric configuration and highlight the effectiveness of finite element analysis for predicting and optimizing origami-inspired metamaterial behavior.

Background and Motivation:

Origami-inspired mechanical metamaterials offer a powerful route to achieving large, programmable deformations, tunable stiffness, and nonlinear load–displacement responses from simple geometric principles. Among these structures, the Kresling unit cell is of particular interest due to its intrinsic coupling between axial compression, torsion, and buckling-driven shape transformations. This strong geometric nonlinearity enables functionalities such as multistability, energy absorption, and mechanically encoded actuation, which are difficult to realize using conventional solid mechanics alone, so we wanted to model through FEA (Finite Element Analysis).

In order to learn more about how we can utilize materials as the driving forces behind mechanical mechanisms, based solely on properties as opposed to a reliance on power transmission and electrical systems, we want to create an accurate model of a Kresling unit cell to further the development of metamaterials relying on the precise geometry and pressures within the unit cell [1]. Having an accurate representation of this compression makes it easier to use these mechanical systems in different applications under varying compressive conditions. Our motivation also stems from having the intellectual curiosity to want to apply the concepts we have learned so far in class to a model that has very complete geometry in conjunction with many different physical properties that contribute to a unique behavior so we can test the limitations of abaqus and see where the failure points populate. We choose to model a plastic based polymer physical model so we can also begin to understand how plasticity, elasticity springs, material compliance, all culminate in a complex behavior pattern and how this changes over a cycle of compressions.

Design Process:

1.1 Design Assumptions

Before the design of the Kresling unit cell was started in ABAQUS or other software, we began with initial assumptions regarding how we would be modeling the unit cell. These assumptions are as follows:

- 1) The different facets of our model are represented by shell parts.

- 2) The creases along every edge of the origami model are represented and assumed to behave as torsional springs.
- 3) Our top polygonal faces are assumed to be rigid bodies
- 4) The available materials for construction of the physical model are ABS and PLA 3D-printing filament.

1.2 CAD Model

1.2.1 CAD Model Importation into Abaqus CAE

Due to the complexity of the geometry for this model, the initial plan was to fully model the Kresling unit cell in Solidworks, and then import the STEP file into Abaqus in order to avoid using the more complicated interface in Abaqus to create the complicated geometry. The primary reason this was an issue was due to the assumptions of the model. In Solidworks CAD software, a part can not be assumed as a “shell”, so the imported geometry did not represent what we were trying to model in Abaqus. There was unrealistic overlap in the corners, where the material went through itself, which made it impossible to model the torsional springs we were aiming for, and also made it a poor representation of our model, both in the CAD assembly and in Abaqus.

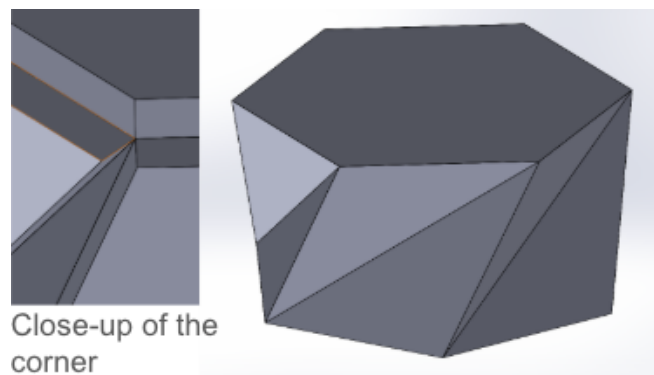


Figure 1: (left) Blown-Up View of the Corner Issues for the CAD Model, (right) Solidworks CAD Model of a Kresling Unit Cell

1.2.2 3-D Printing a Physical Model for Design Validation

In order to 3D print our model, we changed the design in Solidworks from representing what it looks like post-completion to being a panel that we could print, with half thickness portions at the fold lines, such that it would be easy to crease and fold the design. We 3D printed the unit cell as three separate pieces. The folds between the triangular facets were the thinner material creases, and between the hexagonal top and bottom pieces and the triangular geometry we added tape. The model was printed several times, with our initial design having one-inch sides and being made from PLA. The issues with this model were that the PLA was too brittle and plasticized too quickly, and that the one-inch sides were difficult to model in Abaqus as the metric system is preferred. For the final design we used ABS, and printed our designs with side lengths of four centimeters.



Figure 2: 3D Printed Physical Model of the Kresling Unit Cell

1.3 Abaqus Assembly Method

The next steps to creating the model in Abaqus was to create two parts- a triangular facet and a hexagon representing the top and bottom of the unit cell. A third part was then made that contained both the top and the bottom surfaces, which was done by mirroring the original hexagon across a datum plane. The full model was put together as a unit cell in the assembly module, where edge-to-edge and coincident point constraints were used to orient the first two triangular facets with the hexagons. Following that, the two triangles were patterned radially with respect to the center of the hexagon.

This ended up being a successful way of modeling- and with this model we were able to have successful simulations. However, this model was difficult to parametrize, and the lengths of the triangles that were used to define the model were hard to make an equation to represent. In order to get the lengths of the sides of the triangles so that we could define the parts when sketching, we used the measure tool in the CAD that we had.

1.4 Abaqus Part Method

To fix the parametrization issue, we changed the way the part was modeled entirely. As opposed to the traditional method of creating a shell element via a sketch, we made the shell elements using the “Cover Edges” feature in Abaqus. To do this, we created datum points that represented the vertices of both hexagons, and then created wire features that connected the datum points together in the desired shape- this being either a triangular facet or polygon. In this way, we were able to parametrize by the height of the model, the radius of the polygons, and the number of sides of the polygon. This gave us a lot of freedom regarding the parametrization of our model, and allowed us to sweep over various geometry in order to find optimization across these parameters.

Each location of a datum point in the model was defined by four input parameters, N , R , H , and α and two derived variables, θ and ϕ , which were defined as follows:

$$\begin{aligned} N &= \text{Number of Side of the Polygon} \\ H &= \text{Height of the Unit Cell} \end{aligned}$$

R = Radius of the Cylinder Tangential to Both Polygons
 α = Twist Between Top and Bottom Hexagon (in Degrees)

$$\theta = \frac{2\pi i}{N}, i = [1, N - 1]$$

$$\phi = \alpha \frac{\pi}{180}$$

For each datum point on the bottom hexagon, the points were located at:

$$x = R \cos(\theta)$$

$$y = R \sin(\theta)$$

$$z = 0$$

For each datum point on the top hexagon, the points were located at:

$$x = R \cos(\theta + \phi)$$

$$y = R \sin(\theta + \phi)$$

$$z = H$$

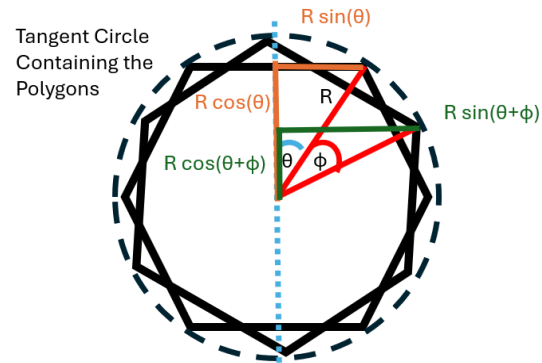
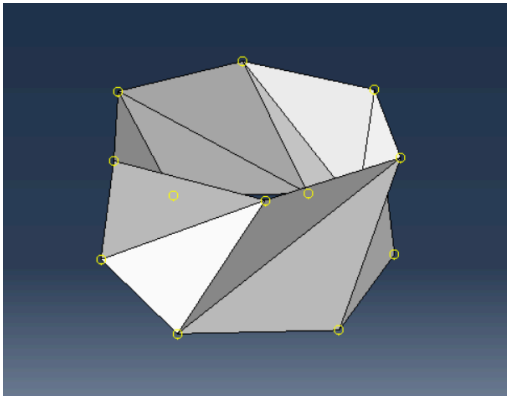


Figure 3 (Left) Datum Point Driven Part Creation in Abaqus CAE, Figure 4 (right) Geometric Figure representing how the coordinate locations are derived

1.5 Material Properties

Initially for our material, the entire model has PLA applied as the material. Due to the issues with PLA in the physical model, we switched from PLA to ABS. We also changed how the material was sectioned. Initially, in order to enforce the rigidity of the top and bottom hexagon we applied rigid body constraints, but this caused there to be compliance issues in the model as certain aspects of the model were overconstrained with these constraints applied. To amend this issue, we added steel material properties to our material library, and applied these conditions to the top and bottom hexagons. This ensured the hexagons were much more rigid than the ABS triangular facets, but it did not enforce them to be entirely rigid, which fixed the over-constraints in our model but allowed them to continue with the assumptions that the top and bottom hexagons are approximately rigid bodies.

Table 1 Material Values			
	PLA	ABS	Steel
Young's Modulus (Pa)	2.58e9	2.08e9	200e9
Poisson Ratio	0.35	0.35	0.3

1.6 Hinge Connections

1.6.1 Different Ways to Model a Torsional Spring Connection with ABAQUS CAE

Multiple spring modeling approaches were evaluated to identify a method capable of representing hinge behavior oriented off the global coordinate axes. Initial simulations using connector elements between edges resulted in immediate snapping into a folded configuration, failing to reproduce the smooth compression observed experimentally. These simulations were performed on a fully merged model; however, the resulting spring behavior was difficult to quantify or control, making this approach unsuitable.

The modeling approach was subsequently revised to use spring–dashpot elements, which allow explicit specification of linear and rotational stiffness as well as controlled rotational freedom. A rotational locking parameter was also introduced to limit excessive rotation and prevent panel interpenetration. This method provided improved numerical stability and greater control over hinge behavior.

1.6.2 Node-to-Node Spring Element Connections for Shared Edges

The modeling technique was changed to tolerate different mesh sizes without changing our spring constant and better control our physical properties. This was done by un-merging the model on the assembly level and defining a spring-connection between existing nodes that occupy the same space. Node sets of all the nodes defined by the seed size along each coincidence edge were created, and then spring dashpots were coded to exist in between every corresponding node for each edge [5]. This code for the springs is based on Leon Kamp's original code segment. To correctly orient these springs to be torsional along many different edges, datum axes were individually created for every spring and the springs were constrained to rotate about this new axis. There are both linear and rotational spring components; the linear component was set to be arbitrarily high so the un-merged faces would not displace away from each other at high loads. The rotational spring constant was set low for initial model validation and then the exact number was determined experimentally. This spring set up allowed enough compliance for this model to fold correctly while maintaining integrity and physical meaning.

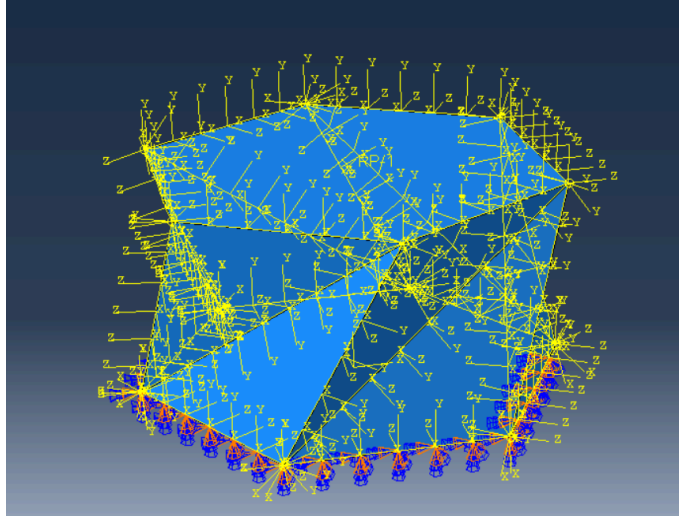


Figure 5: Datum Axes for Every Spring Connection Between two Nodes and Encastre on the Bottom Plane of the Model

1.7 Boundary Conditions

For the boundary conditions, the bottom polygonal face was encastred, and a reference point coupled to the top polygonal face was displaced in Z. This gave our model the ability to freely rotate only through the face being compressed which mimicked a compressive force done in the physical model testing via an Instron compression test. Other attempts to pin the top face lead to aborted simulations due to over-constraints in the system, and any attempts of making the facets rigid bodies led to similarly failed simulations. A load force instead of an applied displacement did not allow us to get an accurate spring constant as the Instron test we ran compressed to a set distance that we could then easily replicate in our simulation using the exact dimensions.

1.8 Steps

For our step, we did not venture outside of the static general solver since our simulation is quasi-static and we were not having any convergence or runtime issues. However, we were experience issues with our simulation aborting or snapping to a configuration immediately, and to resolve this we turned NLGeo on and also added energy dissipation.

Initially, the energy dissipation added in order to allow the automatic stabilization of the model was $2 \cdot 10^{-7}$, as at very high force areas this small dissipation still allowed the solution to reach the full physically possible range of compression, and even go beyond what was possible with the physical model with no convergence issues, though this did not remain the case when we considered adding plasticity to the model. This energy dissipation, although not a physically accurate addition, did not take away from the accuracy of our model because it was negligible in magnitude. In order to find the minimum necessary amount of energy dissipation for the simulation to run, we tested values on a range of $2 \cdot 10^{-7}$ to $2 \cdot 10^{-1}$ and saw that $2 \cdot 10^{-4}$ maintained the same reaction forces as the aborted test pictures with

smaller energy dissipation. The larger values, $2 \cdot 10^{-2}$ and $2 \cdot 10^{-1}$, began to be too different from the actual model as seen how the force curve shifted significantly lower.

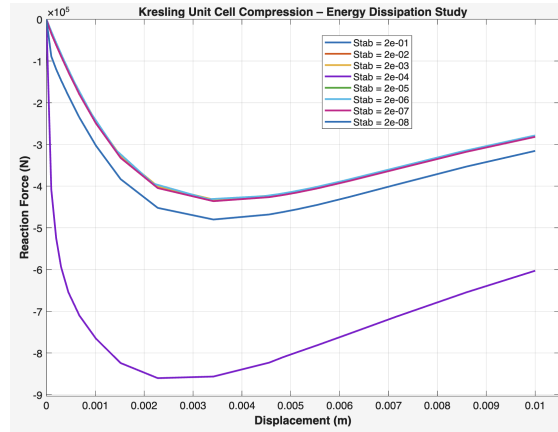


Figure 6: Energy dissipation sweep

Post Processing:

2.1 Calibrating the Spring Constant with Physical Testing

To find our model's spring constant we took our experimental data from the Instron where we did a compression test with one side of our model fixed to a compression plate and the other free to rotate and it was compressed 10 millimeters while the force was being measured. We used a physical model with a height of four centimeters and a radius of four centimeters, and matched these parameters in our simulation. We cycled our physical model 10 times to see the effect of plasticity as well. Then through trial and error, matched the correct spring constant to the model. To determine the spring constant we took the slope of the force vs displacement data and divided that by the twelve edges and compared the reaction force of the physical model to the simulation. This process is detailed more below, and we found spring constants of 885N/m and 905 N/m for the compression of our simulation.

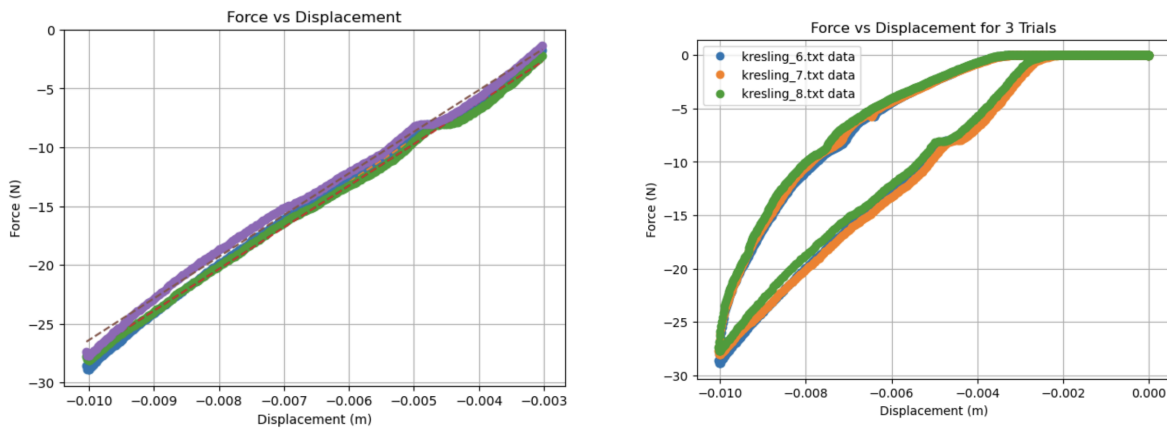


Figure 7 (left): Force v. Displacement curve of Simulation Data that approximates the Physical Model

Figure 8 (right): Force vs. Displacement curve of the physical model

The slope of the force vs. displacement graph can be calculated by:

$K = \text{Spring Constant}$

$F = \text{Applied Force}$

$X = \text{Axial Displacement}$

$B = \text{offset accounting for the delayed compression start point}$

Hooke's Law: $F = Kx + B$

A least squares linear regression was applied on every datapoint while the model is compressed in order to determine an effective axial spring constant for the Kresling unit cell. This approach accounts for geometric nonlinearity, material compliance, and experimental noise by fitting a best-fit linear response across all data points. The resulting slope provides a representative stiffness value that enables consistent and objective comparison between experimental measurements and finite element simulation results.;

$$F = \frac{\sum_{i=1}^n (X - \bar{x})(F - \bar{F})}{\sum_{i=1}^n (X - \bar{x})^2}$$

2.1.2 Plasticity and Hysteresis

Accurately replicating the plastic behavior observed in the physical model proved challenging, suggesting that mechanisms beyond material plasticity alone contribute significantly to the system response. When ABS plasticity properties were introduced into the simulation, excessive permanent deformation occurred after a single compression cycle, which did not match experimental observations, as shown in Figure 9.

An attempt was also made to introduce elastoplastic behavior directly into the spring elements; however, no suitable formulation or setting was available within Abaqus to support this approach. This limitation motivated the consideration of alternative modeling strategies, which are discussed in the Future Considerations section, to more accurately capture the observed hysteresis and plastic effects.

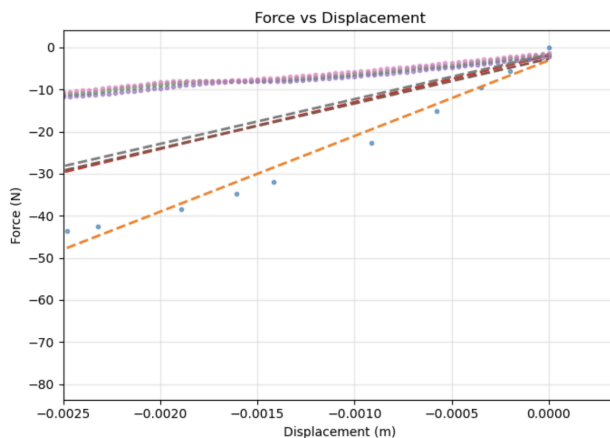


Table 2: Plasticity Parameters

	Value (MPa)	Percent (%)
Yield Point	45	0.0
Hardening	50	0.26
Near Ultimate	55	0.736
Softening	57	1.784

Figure 9: (left) Plasticity overtime of our physical model and the behavior of our simulation with ABS plasticity, Table 2: (right) Plasticity Parameters applied in our model

2.2 Geometry Parametrization

To evaluate the influence of geometric parameters on the mechanical response of the Kresling unit cell, a series of parameter sweeps was conducted. Force–displacement curves obtained from these simulations were analyzed to assess how geometric variation affects stiffness and load-bearing capacity, with the goal of identifying trends that could inform geometric optimization.

2.2.1 Changing the Number of Sides

The first parameter sweep examined the effect of varying the number of polygon sides, with values ranging from 5 to 10. As the number of sides increased, the maximum force sustained by the structure also increased, indicating enhanced stiffness and load resistance. This trend continued up to an 8-sided polygon, at which point the structure exhibited its highest force capacity. For higher side counts, the maximum reaction force decreased, suggesting that additional sides reduced structural effectiveness due to changes in deformation mode and load distribution.

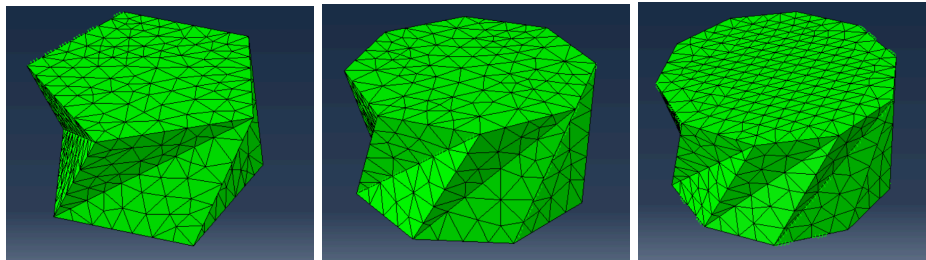


Figure 10: (Left) 5-Sided Polygon, (Middle) 8-Sided Polygon, (Right) 10-Sided Polygon

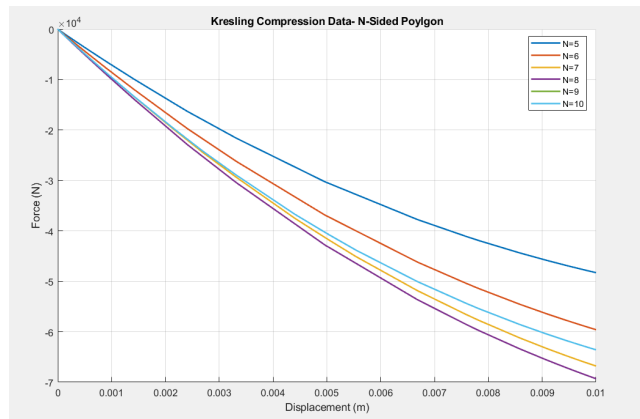


Figure 11: Force-Displacement curves for different N-Sided Polygon Unit Cells

2.2.2 Changing the Ratio of the Radius of the Polygon to the Height of the Structure

The next parameter sweep examined how the relationship between the radius of the tangential circle containing the polygons and the height of the unit cell affected the force response. It was found that the force response decreased as the ratio increased. The ratio was defined as the radius divided by the height, and for this sweep we swept from a radius of two centimeters to a radius of eight centimeters while maintaining the four centimeter height. As the size of the polygon increased relative to the height of the unit cell, the force response became weaker, so geometrically, a proportionally taller unit cell demonstrates greater load-bearing capacity than a wider configuration.

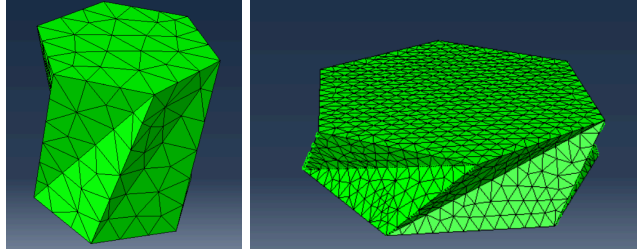


Figure 12: (Left) Geometrically Tall Model (Ratio 0.5), (Right) Geometrically Wide Model (Ratio 2.0)

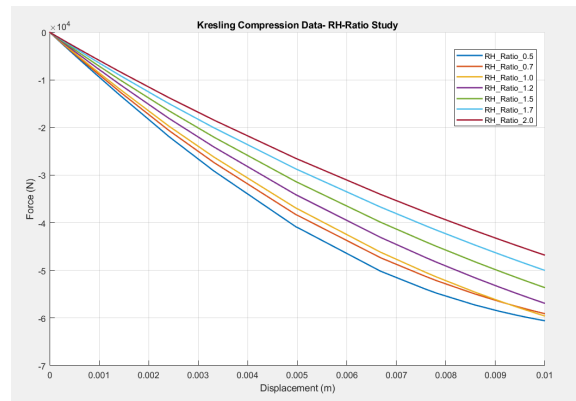


Figure 13: Force-Displacement curves for different Radius to Height Ratio

Conclusions and Takeaway:

An accurate geometric representation of the Kresling unit cell was successfully developed, and a complete workflow for experimental validation of a finite element model was established. The study demonstrated that material behavior plays a critical role in governing the mechanical response of origami-inspired structures, with contributions from elasticity, plasticity, friction, and geometric nonlinearity. Finite element simulations proved valuable in isolating and analyzing these effects, significantly accelerating the process of understanding the forces driving the observed behavior. The results also highlighted the sensitivity of the force–displacement response to experimental setup and boundary conditions, indicating that further refinement of testing procedures could improve data fidelity.

Although the calibrated model successfully matched the effective spring constant measured experimentally, the unloading behavior did not accurately reproduce the hysteresis observed in physical testing. Specifically, the simulated response exhibited an elastic return that closely followed the loading

path, suggesting that additional mechanisms beyond the current spring formulation are required to capture elastoplastic behavior more accurately.

Future Considerations:

Future work could extend the optimized Kresling unit cell model to application-driven studies, including investigations of pumping behavior and pressure-handling capability across varying geometries. The effects of stacking multiple unit cells in series could also be explored, as coupled origami structures often exhibit significantly different mechanical responses compared to single units.

Additional modeling strategies for crease behavior warrant investigation. One potential approach involves replacing spring-based hinge representations with filleted or locally thinned regions along fold lines, allowing hinge compliance to emerge directly from material properties rather than prescribed spring elements. While such methods may introduce additional compliance challenges, they could provide a more physically representative model of elastic and plastic folding behavior and capture deformation mechanisms not accessible through discrete spring connections.

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